Sabha

Conversation Intelligence Platform

Project progress report: everything built so far

As of	10 November 2026 (end of pilot week 4 of 8)
Work started	14 October 2026
Previous report	27 October 2026
Code branch	week1 (182 commits), pushed to GitHub on 10 November
Pilot scope	Teaching and reception in schools; sales meetings, support calls, team meetings and company reception added for companies
Status	Full pipeline working end to end on a developer machine, on test recordings. Main screens redesigned to the v2 design

1. Summary

Sabha turns recorded conversations (classroom lessons, front-desk calls, sales meetings, customer support calls, team meetings) into a transcript that says who spoke when, and a feedback report that scores the staff member against a rubric. Every comment in the report is backed by a quote from the conversation with its timestamp, and software checks each quote really appears in the transcript before anyone sees it.

Since the last report on 27 October, the work has been about making the product ready to show and to sell:

- **A redesigned web app (UI v2).** New look across the main screens: Overview dashboard, Recordings list, the lesson and call report, a new **My progress** home page for staff, and Upload.
- **Customer support calls**, a sixth conversation type for companies: the agent is scored on helping the customer, and a case card records the product, problem, steps tried, outcome and follow-up.
- **Cost per recording in rupees**: an expected cost at upload and the actual cost when done, from stored records.
- **Sharing a report** with the person analysed, with the manager's note; **meeting transcripts** (Google Meet) can be uploaded instead of audio.
- **Better speaker detection** on phone calls, **classes and sections** for schools, nightly database backups with a restore drill, and many smaller fixes.

182	~27,000	~19,000	~1,510
commits (130 on 27 Oct)	lines of Python (app)	lines of web app code	automated tests

Counted from the source code the same way as last time: about 1,256 Python test functions plus about 254 web unit tests. Lint and tests run locally before every commit (there is no CI server, ADR 0010).

Where it stands honestly. Everything runs on a developer machine on test recordings. It has not yet run on real school data, and it is not deployed to AWS. There are no accuracy figures yet (word error rate, AI-to-human agreement): the evaluation tools are built, and figures will come only from running them on real, human-checked recordings. No such number appears in this report on purpose. The new rubrics for companies are drafts waiting for the owner's review.

2. How Sabha works

Step	What happens	Built
1. Upload	The browser sends audio straight to secure storage in parts (large files, resumable). The server only authorises it and records it. Checksums catch corrupt files; duplicates are detected. **New:** a Google Meet transcript (.txt) can be uploaded instead of audio (ADR 0009); expected cost shown before upload.	Yes
2. Preprocess	ffmpeg checks and normalises the audio. Too-short or too-long files get a clear message.	Yes
3. Silence detection	Silero VAD removes silence for faster processing, but keeps a map so every timestamp shown is on the original recording's timeline.	Yes
4. Who spoke when	Speaker diarisation with pyannote (CPU). **New:** support calls are treated as exactly two speakers, one at a time. NeMo Sortformer is planned once a GPU exists.	Yes (CPU)
5. Transcribe	Speech to text through a provider layer (Sarvam AI). **New:** no speech chunk crosses a change of speaker, words stay with the person who said them, and when Sarvam is busy the chunk is retried and the app shows "retrying". Speaker roles on support calls come from what each person does (ADR 0011); unclear cases go to human review.	Yes
6. Analyse	Numbers (talk share, questions, pace, silence, longest monologue, hold time) are computed in code. The AI only judges rubric dimensions and must quote evidence; a verifier checks each quote.	Yes
7. Report	Scores, evidence quotes with timestamps, strengths and next steps, plus type-specific cards (reception ticket, sales deal card, support case card, meeting notes). Actual cost in rupees. Email when ready; live status in the app.	Yes

Each stage runs on its own work queue (Celery on Redis), retries safely, and skips work that is already done, so a crash or restart never produces duplicate results. Failed jobs land in a dead-letter queue and can be retried or closed from the app.

3. Conversation types and rubrics

Type	Rubric file	What the report gives
School teaching	teaching.school.v0.1	Teacher scored on rubric dimensions with quotes; talk share, questions, pace, silence, monologue length. No student is scored.
School reception	reception.school.v0.2 (reviewed)	Receptionist scores, a call ticket, hold time; dimensions that do not apply are marked N/A.
Company reception	reception.company.v0.1 (draft)	Walk-in visitor at a business front desk; own ticket categories; ticket backed by two verified quotes.
Sales meeting	sales.company.v0.3 (draft)	In-person meetings and online demos (also from Meet transcripts). Rep scored on the client's sales method plus best practice. Deal card: need, setup, objections, interest, next step, follow-up; every item quote-backed. v0.3 changes weights only.
Support call **(new)**	support.company.v0.3 (draft)	Phone calls between a support agent and a customer. Scored on helping the customer, not call etiquette. Case card: product, problem, steps tried, outcome, follow-up. Switched on for every company.
Team meeting	meeting.company.v0.1 (ADR 0008)	The meeting is scored, not any person: goal, decisions, actions, everyone heard. Notes with action items, owners and dates.

Companies now use their own words: "Team member" for people, "Sales rep" on sales calls and "Agent" on support calls. A demo company (Brightpath Software) is seeded for showing sales, support and meetings.

4. The web app: UI v2 redesign

The main screens were rebuilt to the v2 design in docs/design/v2: one blue for the brand and the staff speaker, the Geist font, score colours by band, a "who spoke when" strip on every recording, and a header with search and a notification bell. Words such as "Teacher" or "Lesson" still come from each organisation's settings, never from the code.

Page	What it does	v2 look
Overview	Greeting, campus switcher and period; tiles with eight-period trends; "Needs your decision"; "Processing now" with expected ready time; where coaching will help most, by campus and week; recent recordings.	Done
My progress **(new)**	Home page for staff (teachers, sales reps, support agents): their own numbers, newest report with the manager's note, what went well and what to try next, the three rubric points to work on, recent recordings. No comparison with colleagues.	Done
Recordings	Tabs with counts; filters by type, class, person, date and score; select several to re-analyse or share; retry or close failed ones; CSV export.	Done
Report	Coaching steps for managers; two-lane timeline with zoom; rubric levels; transcript beside the scores with synced audio; cost and processing under Technical details; share with the person.	Done
Upload	Type cards with colours and what each is scored on; status per file; expected cost; audio hours for owners and admins; consent card.	Done
People	Teachers with the classes and sections they teach (schools: LKG and UKG before Class 1 to 12).	Partly
Reviews	Speaker questions, not-usable recordings and disputed scores.	Next
Sales report	Deal card today; the v2 design adds a meeting-steps checklist and time to first demo.	Next
Rubrics, Settings, Activity	Read-only rubrics; organisation settings showing only usable types; audit log with grouped views.	Styled

5. Roles and permissions

Role	Can do
Owner	Everything, including inviting any role, settings, erasure and re-analysis.
Admin	Run the organisation; invite managers, reviewers and staff; re-analysis; activity log; plan usage.
Manager	See and upload for people in their own units; share reports; resolve reviews and disputes for those units.
Reviewer	Like managers for their units: review queue and dispute decisions.
Staff	Upload for themselves; see their own reports and My progress; raise disputes. Never resolve their own.

Sharing a report emails the person a link and shows the manager's note; it does not change who may see the report. Every API route is in a permission matrix, and a test fails if a new route is missing.

6. Trust, privacy and safety: how the rules are enforced

Rule	How it is enforced in the code
Customer data kept apart	Every record carries the organisation id; all reads and writes go through one tenant-scoped repository that refuses a query without it. A security test fails if any code touches the database directly.
No made-up claims	Numbers are computed in code. Each AI judgement must cite quotes; a verifier confirms each quote is in the transcript (fuzzy match 0.90 or better). Unverified output is never shown. Costs come from stored records and published prices.
Personal data masked	Phone numbers, Aadhaar-like and PAN-like numbers, emails and student names are masked before any text goes to the AI. Evidence quotes are shown redacted to everyone.
Students not analysed	Only staff are scored. Meetings score the meeting, not people. My progress shows a person only their own results.
Consent first	A recording without a consent reference cannot move past upload.
No silent fallback	Speech and AI provider switches happen only when configured, and are logged. Sarvam rate limits are waited out and retried, not swapped.
Safe to repeat	Every create accepts an idempotency key; every pipeline stage skips work already done.
Audit trail	Playback, export, sharing, score changes, settings and erasure write an append-only audit log.
Data in India	Configured for AWS Mumbai (ap-south-1) and MongoDB Atlas Mumbai.
Retention, erasure, backups	Audio deleted after the retention period (90 days default). Owners can erase a recording or a person. **New:** nightly database export with a restore drill (locally; Atlas backups come with AWS).
Secrets	Only from environment variables; secret scanning in the commit hooks.

7. Running costs measured so far

From stored records on a developer laptop (no GPU). Single runs on test audio, not averages over real school recordings.

Recording	LLM calls and cost	Diarisation	Transcription	Analysis
28.2 min lecture, 3 Nov, gpt-6-luna	6 calls, $0.0097	about 28 min	46 s	about 2 min, completed

- That run cost **₹12.46** in all, from its cost card: speech recognition ₹11.53 (23.5 of 28.2 minutes sent after silence was removed) and the AI model ₹0.94. Speech recognition is about 93% of the cost.
- Diarisation on CPU takes about as long as the recording itself: it is the main compute cost and the main reason a report takes as long as it does.
- The AI prompt cache did not hit between the two scoring calls; this is an open question.

8. Timeline of what was built

Date	Milestone
14–26 Oct	Covered in the previous report: repository and tooling, the full pipeline, tenant model, upload, AI gateway, rubrics, sales meetings, team meetings, company reception, disputes, re-analysis, live status.
27 Oct	Settings; plan allowances; rubrics pages; role pass and accessibility; audit log; retry and close failed recordings; retention and erasure; **nightly backups and restore drill**. UI v2 begins: design tokens, sidebar, "who spoke when" strips, header search, notification bell, the new Overview.
28 Oct	Cost in rupees (expected at upload, actual when done); local password sign-in; teachers with their classes and sections; security updates.
29 Oct – 3 Nov	UI v2 Recordings page and lesson report; sharing a report with the person; meeting transcript upload (ADR 0009); GitHub Actions removed, checks run locally (ADR 0010); first full cost run.
4 Nov	Support calls: new conversation type, case card, two-speaker diarisation, speaker roles from evidence (ADR 0011); Sarvam rate-limit retries.
5–6 Nov	Support rubric v0.2 and v0.3; company wording (Team member, Sales rep, Agent); transcript and call-report fixes; less noise across activity, settings and overview.
10 Nov	My progress, the home page for staff; Upload page in the v2 style; both pushed to GitHub.

9. Technical overview

Area	Choice
Backend	Python 3.12, FastAPI, Pydantic v2; async MongoDB (PyMongo); uv workspaces; ruff; mypy --strict
Workers	Celery 5 on Redis 7; queues cpu.preprocess, gpu.speech, llm.analyze, io.notify, billing.meter; outbox relay and beat scheduler
Database	MongoDB 7 replica set (transactions and change streams)
Storage	S3 (Mumbai) in production; RustFS locally (ADR 0003)
Audio and speech	ffmpeg, Silero VAD, pyannote; Sarvam AI speech recognition
AI	OpenAI gpt-6-luna via our own gateway: routing, budgets, retries, caching, cost per call recorded
Web	React 18, TypeScript, Vite, Tailwind, TanStack Query, React Router, Vitest, Playwright; Geist font
Sign-in	OpenID Connect with Auth0 (ADR 0002); local password sign-in for development
Checks	Lint, type checks and tests run locally before every commit; no CI server (ADR 0010)

API: 46 operations under /v1 (counted from the running API's OpenAPI document), including the new inbox, sharing, CSV export, cost estimate and cost per recording. Errors follow RFC 9457 with a stable code and request id; lists use cursor pagination.

Architecture decisions recorded (ADRs): 0001 repository and stack, 0002 Auth0, 0003 RustFS, 0005 fair GPU scheduling, 0006 Sarvam AI, 0007 leaving "Needs review", 0008 meetings scored as a whole, **0009 transcript upload, 0010 no GitHub Actions, 0011 evidence-based speaker roles** (new since the last report).

10. Remaining work

Product work

Item	What remains
UI v2	Reviews page; sales report (meeting-steps checklist, time to first demo, Add to CRM); a small fix so pages never scroll sideways on phones
Rubric reviews	Owner's review of the draft rubrics: teaching v0.1, company reception, sales v0.3, support v0.3, team meetings
Rubric editing	Editing and publishing rubrics in the app (needs calibration with real rater scores); choosing a rubric version for re-analysis is in the API only
Plans and billing	Billing system; usage export for invoicing
Webhooks	Webhook settings and delivery (signing is built; emails are done)
Limits	Rate limiter in the sign-in step; the 80% usage warning email
Admin tools	Pause processing for one organisation; admin screens for stuck jobs

Quality work (needs real, human-annotated recordings)

- AI model bake-off; calibration of AI scores against human raters; an evaluation gate before releases.
- Diarisation on GPU (NeMo Sortformer) once a GPU is available.
- Quality gate reviews 1–3 against real numbers.

Cost questions

- Why the prompt cache does not hit, and whether caching is worth it with gpt-6-luna.
- GPU or cheaper machines for diarisation; a Sarvam plan versus pay as you go; expected pilot volume.

Infrastructure (planned together in week 8)

- AWS staging and production (Docker Compose, Terraform), Auth0 tenant and production sign-in, deployment.
- Monitoring: Prometheus, Grafana, Sentry, alerts; load and chaos tests; security review.
- Onboarding pack, go-live, daily health report.

Figures in this report are counts from the repository on 10 November 2026 (commits, lines, test functions) and from the running API (endpoints); costs are quoted from stored records. No quality or accuracy figures are claimed.